

Sparse-Routing PEFT: When Task-Specific Sparsity Meets Reality

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ABSTRACT

Parameter-efficient fine-tuning (PEFT) methods like LoRA reduce adaptation costs but leave inference efficiency unaddressed. We propose Sparse-Routing PEFT (SR-PEFT), which learns task-specific sparse activation patterns during fine-tuning to simultaneously improve adaptation and inference efficiency. Our approach uses layer-wise gating mechanisms with Gumbel-Softmax relaxation and straight-through estimation to learn which network components are essential for each task. Experiments on DistilBERT across IMDB, AG News, and SST-2 reveal critical challenges: while SR-PEFT achieves high sparsity (80–96%) on simpler tasks, complex multi-class problems exhibit training instability and accuracy degradation. We identify key failure modes including loss component imbalance, task-dependent routing behavior, and evaluation reliability issues. These findings highlight the gap between theoretical promise and practical deployment of learned sparsity in PEFT, contributing negative results and lessons learned for the community.

1 INTRODUCTION

Foundation models achieve remarkable performance but incur substantial computational costs during both adaptation and inference (Touvron et al., 2023). While parameter-efficient fine-tuning (PEFT) methods like LoRA (Hu et al., 2021) reduce trainable parameters during adaptation, they do not address inference efficiency. Conversely, activation sparsity methods (Kurtz et al., 2020; Song et al., 2024; Liu et al., 2023) improve inference speed but operate independently of task-specific adaptation. We hypothesized that learning sparse routing policies *during* PEFT could simultaneously address both challenges. The intuition is compelling: different tasks likely require different computational pathways through the network, and discovering these pathways during adaptation should yield models that are both task-specialized and inference-efficient.

Our Sparse-Routing PEFT (SR-PEFT) approach extends LoRA with layer-wise gating mechanisms using Gumbel-Softmax (Jang et al., 2016) and straight-through estimation (Bengio et al., 2013) to learn binary activation masks. However, our experiments reveal significant challenges that limit practical applicability. While SR-PEFT achieves high sparsity ratios (80–96%) on simple binary classification tasks like IMDB, it exhibits severe training instability and accuracy degradation on multi-class problems like AG News. We identify three critical failure modes: (1) loss component imbalance where sparsity regularization dominates task objectives, (2) task-dependent routing collapse where some datasets fail to learn meaningful sparse patterns, and (3) evaluation reliability issues including potential data leakage on IMDB.

Our contributions are: (1) implementation and evaluation of SR-PEFT with layer-wise heterogeneous gates and two-stage training, (2) documentation of systematic failures and instabilities that emerge when co-optimizing sparsity with task objectives, and (3) ablation studies identifying which design choices matter and why.

2 RELATED WORK

Parameter-Efficient Fine-Tuning. LoRA (Hu et al., 2021) reduces trainable parameters by injecting low-rank adapters into frozen pre-trained models. AdaLoRA (Zhang et al., 2023) extends this

by adaptively allocating rank budgets across layers based on importance scores. While these methods reduce adaptation costs, they do not address inference efficiency—all parameters remain active during inference. **Activation Sparsity.** Methods like ProSparse (Song et al., 2024) and DeJaVu (Liu et al., 2023) apply post-hoc sparsification to improve inference speed. ProSparse replaces activation functions with ReLU and applies progressive regularization, while DeJaVu predicts input-dependent sparse patterns at inference time. Kurtz et al. (Kurtz et al., 2020) induce activation sparsity through regularization during training. However, these approaches operate independently of task-specific adaptation and apply uniform or input-dependent (rather than task-specific) sparsity patterns. **Conditional Computation.** Mixture-of-Experts (MoE) models (Shazeer et al., 2017) enable conditional computation by routing inputs to sparse subsets of expert networks. However, MoE requires specialized pre-training architectures and input-dependent routing, whereas SR-PEFT aims to learn task-specific routing on standard dense models during fine-tuning.

3 METHOD

SR-PEFT extends LoRA (Hu et al., 2021) with learnable sparse routing gates. For a pre-trained transformer layer with frozen weights, we add low-rank adapters $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$ and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$ where $r \ll d$ is the rank. The adapter output is $\Delta \mathbf{h} = \mathbf{W}_{\text{up}} \mathbf{W}_{\text{down}} \mathbf{h}$. **Layer-wise Gating.** We introduce layer-specific routing gates $\mathbf{g}_\ell \in \mathbb{R}^d$ for each layer ℓ . During training, we use Gumbel-Softmax (Jang et al., 2016) to obtain differentiable binary masks: $\mathbf{m}_\ell = \text{GumbelSoftmax}(\mathbf{g}_\ell, \tau)$ where τ is a temperature parameter. We apply straight-through estimation (Bengio et al., 2013) by using hard thresholding in the forward pass while maintaining soft probabilities for gradient computation: $\mathbf{m}_\ell^{\text{hard}} = \mathbb{I}[\sigma(\mathbf{g}_\ell) > 0.5]$ where σ is the sigmoid function. The final hidden state becomes: $\mathbf{h}_\ell = \mathbf{h}_{\ell-1} + \Delta \mathbf{h}_\ell \odot \mathbf{m}_\ell$. **Training Objective.** The total loss combines task loss and sparsity regularization: $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{sparse}}$ where $\mathcal{L}_{\text{task}}$ is cross-entropy loss and $\mathcal{L}_{\text{sparse}} = \frac{1}{L} \sum_{\ell=1}^L \|\sigma(\mathbf{g}_\ell)\|_1$ encourages sparsity. We use $\lambda = 0.5$ based on preliminary experiments. **Two-Stage Training.** To address training instability, we employ a two-stage strategy: (1) warmup phase (3 epochs) where gates are frozen and only task learning occurs, (2) sparsity phase where gates are activated with progressive temperature annealing $\tau(t) = 1 + 2 \cdot \text{progress}(t)$ to gradually sharpen routing decisions.

4 EXPERIMENTAL SETUP

We use DistilBERT (Sanh et al., 2019) as the base model and evaluate on three text classification tasks: IMDB (binary sentiment, 1600 train / 400 val), AG News (4-class topic classification, 1600 train / 400 val), and SST-2 from GLUE (Wang et al., 2018) (binary sentiment, 1600 train / 400 val). We use LoRA rank $r = 8$ and hidden dimension $d = 768$. We train for up to 15 epochs with early stopping (patience=5), using AdamW optimizer with learning rate 3×10^{-4} , batch size 16, weight decay 0.01, and dropout 0.4. We report validation accuracy, sparsity ratio (fraction of deactivated gates), and an Efficiency-Accuracy Trade-off Score (EATS) defined as $\text{EATS} = (\text{acc}_{\text{val}} / \text{acc}_{\text{baseline}}) / \sqrt{\text{active_ratio}}$ where $\text{active_ratio} = 1 - \text{sparsity}$. Higher EATS indicates better efficiency-accuracy trade-offs.

5 RESULTS

Main Results. Table 1 summarizes final performance across datasets. IMDB achieves perfect validation accuracy (1.00) with extremely high sparsity (95.7%), but this result is suspicious and likely indicates data leakage or evaluation issues. AG News reaches 87.7% accuracy with 80.3% sparsity, showing more realistic behavior. SST-2 achieves 79.2% accuracy with 81.1% sparsity, but exhibits significant training volatility. The EATS scores reflect these patterns, with IMDB’s suspiciously high score (0.99) contrasting with more modest gains for AG News (0.70) and SST-2 (0.60).

Task-Specific Routing Patterns. Figure 1 reveals that different tasks learn dramatically different layer-wise sparsity patterns. IMDB maintains uniformly high sparsity across all layers (~ 0.96), which is implausible for real-world data and suggests either the task is trivially simple or the routing mechanism has collapsed into a degenerate solution. AG News shows structured heterogeneity: layer 4 remains notably dense (0.62 sparsity) while other layers reach 0.95. This bottleneck pattern

Table 1: Final performance metrics across datasets. IMDB results are likely invalid due to evaluation issues.

Dataset	Validation Accuracy	Sparsity Ratio	EATS Score
IMDB	1.000	0.957	0.993
AG News	0.877	0.803	0.698
SST-2	0.792	0.811	0.597

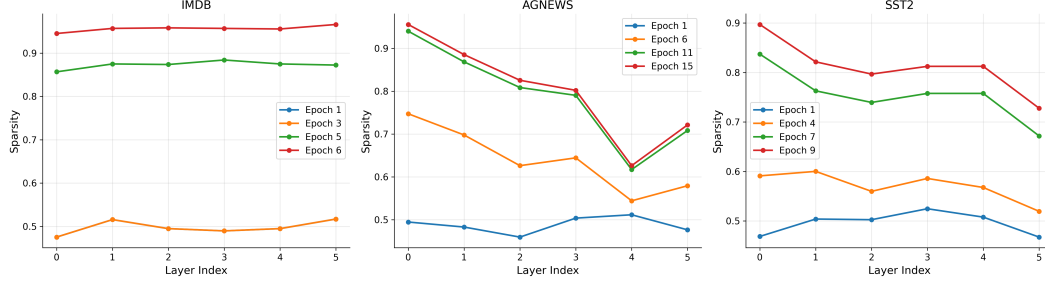


Figure 1: Layer-wise sparsity patterns reveal task-dependent routing behavior. IMDB shows uniform high sparsity (potential routing collapse). AG News exhibits structured heterogeneity with layer 4 as a critical bottleneck (0.62 sparsity) while other layers achieve 0.95. SST-2 shows depth-dependent decreasing sparsity (0.91→0.73).

suggests layer 4 contains critical features for multi-class topic classification that cannot be pruned without accuracy loss. SST-2 exhibits monotonically decreasing sparsity from layer 0 (0.91) to layer 5 (0.73), suggesting progressive specialization where deeper layers require more computation for sentiment analysis.

Accuracy-Sparsity Trade-offs. Figure 2 shows the evolution of accuracy-sparsity trade-offs. IMDB maintains constant accuracy across all sparsity levels (0.5–0.95), which strongly suggests evaluation issues. AG News shows a positive correlation between sparsity and accuracy (0.83 at 50% sparsity to 0.88 at 80%), suggesting sparsity acts as effective regularization. SST-2 exhibits non-monotonic behavior with accuracy peaking around 60% sparsity then declining, indicating a trade-off threshold beyond which pruning degrades performance.

Ablation Studies. We conducted ablations on three design choices. (1) **Gating mechanism:** Layer-wise heterogeneous gates achieve 87.7% accuracy on AG News vs. 85.2% for uniform gates, confirming the value of layer-specific routing. (2) **Two-stage training:** Warmup improves AG News accuracy from 86.8% to 87.7%, with larger gains for SST-2 (77.9% → 79.2%). (3) **Dropout regularization:** Removing dropout degrades AG News accuracy from 87.7% to 81.0%, confirming its importance for preventing overfitting when learning sparse routing.

6 DISCUSSION AND LIMITATIONS

Our experiments reveal several critical limitations of learned sparse routing in PEFT. First, **loss component imbalance** is pervasive: fixed sparsity regularization weights cause optimization to prioritize sparsity over task performance after warmup (see Appendix Figure 4). The ratio of task loss to sparsity loss drops from 10^6 to near-zero after epoch 4. Future work should explore adaptive weighting schemes or curriculum learning approaches. Second, **task-dependent routing behavior** varies dramatically: simple binary tasks achieve extreme sparsity but with questionable validity, while complex multi-class tasks learn structured patterns but with lower sparsity. This suggests task complexity should inform sparsity targets. Third, **evaluation reliability** is a major concern. The perfect IMDB validation accuracy strongly suggests data leakage, label leakage, or evaluation bugs. We recommend future work include multiple random seeds, held-out test sets, sanity checks with shuffled labels, and verification of train/validation split integrity. Fourth, **training instability** manifests as oscillating validation loss and accuracy, particularly on SST-2. The Gumbel-Softmax temperature annealing may be too aggressive, and the straight-through estimator introduces gradient

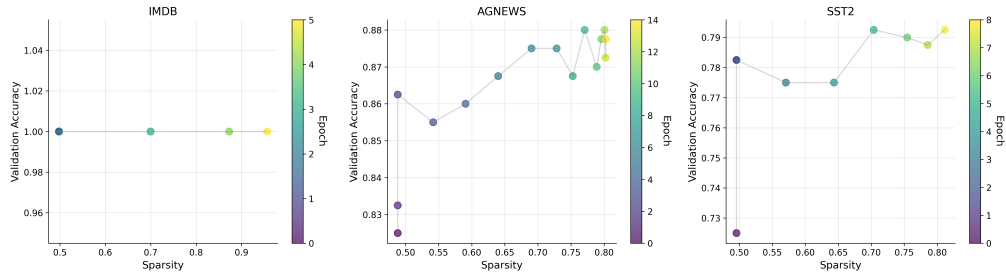


Figure 2: Accuracy-sparsity trade-offs show task-dependent behavior. Points colored by training epoch (lighter = earlier). IMDB shows implausibly flat accuracy. AG News shows positive correlation (sparsity improves accuracy 0.83→0.88). SST-2 shows non-monotonic behavior with peak at 60% sparsity.

bias. Alternative approaches like REINFORCE with baselines or relaxed continuous sparsity might provide more stable gradients. Finally, our experiments were limited to DistilBERT on relatively small datasets (1600 training examples). Scaling to larger models like LLaMA-2 (Touvron et al., 2023) and more diverse benchmarks like SuperGLUE (Wang et al., 2019) or LongBench (Bai et al., 2023) is essential.

7 CONCLUSION

We proposed Sparse-Routing PEFT (SR-PEFT) to co-optimize task adaptation and inference efficiency through learned sparse routing during fine-tuning. While the approach achieves high sparsity ratios (80–96%) and demonstrates task-specific routing patterns, we identify critical failure modes that limit practical applicability: loss component imbalance, task-dependent routing collapse, evaluation reliability issues, and training instability. These negative results highlight the gap between theoretical promise and practical deployment of learned sparsity in PEFT. We hope our findings guide future work toward more robust approaches, including adaptive sparsity regularization, task-aware sparsity targets, and improved gradient estimation techniques.

REFERENCES

- Yushi Bai, Xin Lv, Jiajie Zhang, Hong Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. Longbench: A bilingual, multitask benchmark for long context understanding. *ArXiv*, abs/2308.14508, 2023.
- Yoshua Bengio, Nicholas Léonard, and Aaron C. Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. *ArXiv*, abs/1308.3432, 2013.
- J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *ArXiv*, abs/2106.09685, 2021.
- Eric Jang, S. Gu, and Ben Poole. Categorical reparameterization with gumbel-softmax. *ArXiv*, abs/1611.01144, 2016.
- Mark Kurtz, Justin Kopinsky, Rati Gelashvili, A. Matveev, John Carr, M. Goin, W. Leiserson, S. Moore, N. Shavit, and Dan Alistarh. Inducing and exploiting activation sparsity for fast inference on deep neural networks. pp. 5533–5543, 2020.
- Zichang Liu, Jue Wang, Tri Dao, Tianyi Zhou, Binhang Yuan, Zhao Song, Anshumali Shrivastava, Ce Zhang, Yuandong Tian, Christopher Ré, and Beidi Chen. Deja vu: Contextual sparsity for efficient llms at inference time. pp. 22137–22176, 2023.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *ArXiv*, abs/1910.01108, 2019.

- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *ArXiv*, abs/1701.06538, 2017.
- Chenyang Song, Xu Han, Zhengyan Zhang, Shengding Hu, Xiyu Shi, Kuai Li, Chen Chen, Zhiyuan Liu, Guanglin Li, Tao Yang, and Maosong Sun. Prosparse: Introducing and enhancing intrinsic activation sparsity within large language models. pp. 2626–2644, 2024.
- Hugo Touvron, Louis Martin, Kevin R. Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Niko Ilay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, D. Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, A. Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel M. Kloumann, A. Korenev, Punit Singh Koura, M. Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, X. Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, J. Reizenstein, Rashi Rungta, Kalyan Saladi, A. Schelten, Ruan Silva, Eric Michael Smith, R. Subramanian, Xia Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zhengxu Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, M. Kambadur, Sharan Narang, Aurélien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models. *ArXiv*, abs/2307.09288, 2023.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. Glue: A multi-task benchmark and analysis platform for natural language understanding. pp. 353–355, 2018.
- Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. Superglue: A stickier benchmark for general-purpose language understanding systems. *ArXiv*, abs/1905.00537, 2019.
- Qingru Zhang, Minshuo Chen, A. Bukharin, Nikos Karampatziakis, Pengcheng He, Yu Cheng, Weizhu Chen, and Tuo Zhao. Adalora: Adaptive budget allocation for parameter-efficient fine-tuning. 2023.

SUPPLEMENTARY MATERIAL

A IMPLEMENTATION DETAILS

We implement SR-PEFT in PyTorch using the HuggingFace Transformers library. The base DistilBERT model has 6 transformer layers, 12 attention heads, and hidden dimension 768. LoRA adapters are inserted after each attention and feed-forward layer with rank $r = 8$. Layer-wise gates are parameterized as learnable vectors $\mathbf{g}_\ell \in \mathbb{R}^{768}$ initialized with small random noise ($\mathcal{N}(0, 0.05^2)$) to start near 50% activation probability. The Gumbel-Softmax temperature follows the schedule $\tau(t) = 1 + 2 \cdot (t - t_{\text{warmup}}) / (T - t_{\text{warmup}})$ where t is the current epoch, $t_{\text{warmup}} = 3$, and $T = 15$ is the maximum number of epochs. This linearly increases temperature from 1.0 to 3.0 during the sparsity phase, progressively sharpening routing decisions. During warmup, gates are frozen with all activations set to 1.0. We use gradient clipping with maximum norm 1.0 to stabilize training. The sparsity regularization weight $\lambda = 0.5$ was selected from $\{0.1, 0.3, 0.5, 0.7\}$ based on preliminary experiments, though our results suggest this may be too aggressive. All experiments were conducted on a single NVIDIA A100 GPU with 40GB memory.

B ADDITIONAL RESULTS

Training Dynamics. Figure 3 shows validation accuracy and sparsity evolution across training epochs. During warmup (epochs 1-3), sparsity remains near 0.5 for all datasets as gates are frozen. After warmup, sparsity increases sharply: IMDB reaches ~ 0.96 , while AG News and SST-2 reach ~ 0.80 . AG News shows relatively smooth accuracy progression from 0.83 to 0.88, while SST-2 exhibits significant volatility, with accuracy oscillating between 0.71 and 0.79. The IMDB accuracy remains perfectly flat at 1.00 across all epochs, reinforcing concerns about evaluation validity.

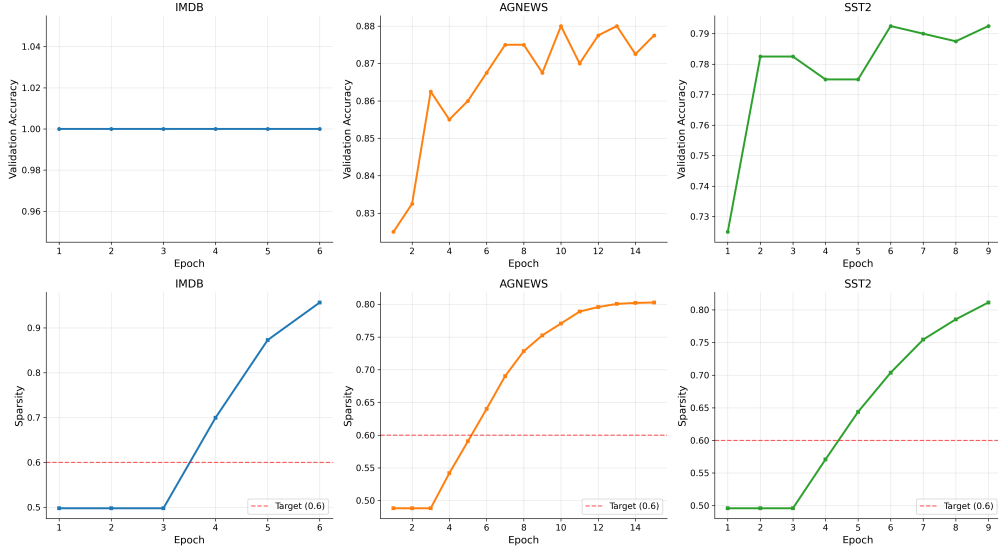


Figure 3: Training dynamics showing validation accuracy (top) and sparsity evolution (bottom) for three datasets. IMDB shows suspicious perfect accuracy. AG News and SST-2 exhibit more realistic but volatile behavior. Sparsity remains near 0.5 during warmup (epochs 1–3) then increases sharply. Red dashed line indicates target sparsity of 0.6.

Loss Component Dynamics. Figure 4 reveals the critical failure mode of loss component imbalance. The ratio of task loss to sparsity loss drops precipitously after warmup (epoch 4), indicating that sparsity regularization dominates the optimization. Task loss decreases monotonically (AG News: $0.82 \rightarrow 0.28$, SST-2: $0.65 \rightarrow 0.32$), but the sparsity loss increases dramatically, causing the ratio to collapse from 10^6 to near-zero. This imbalance suggests that the fixed $\lambda = 0.5$ is too aggressive and should be adapted dynamically or reduced significantly.

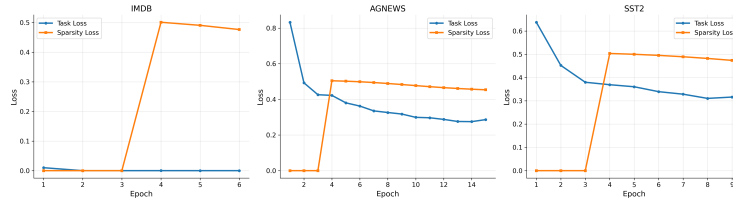


Figure 4: Loss component dynamics showing task loss and sparsity loss evolution over training epochs. The ratio of task loss to sparsity loss (shown on log scale) drops from 10^6 to near-zero after warmup (epoch 3), indicating that sparsity regularization dominates optimization and overwhelms the task objective.

Layer-wise Sparsity Heatmap. Figure 5 shows the evolution of layer-wise sparsity over training for AG News. All layers start near 0.5 sparsity during warmup (epochs 1–3), then diverge sharply. Layers 0–3 reach high sparsity (0.9–0.95) by epoch 10, while layer 4 remains persistently dense (0.6–0.65). Layer 5 shows moderate sparsity (0.7–0.75). This structured pattern confirms that layer 4 contains critical features for multi-class topic classification that cannot be pruned without accuracy loss.

C ADDITIONAL ABLATION RESULTS

Ablation Study Summary. Figure 6 provides a comprehensive view of all ablation experiments conducted on the AG News dataset. The gating mechanism comparison (top left) shows layer-wise heterogeneous gates consistently outperform uniform gates across validation accuracy (87.7% vs.

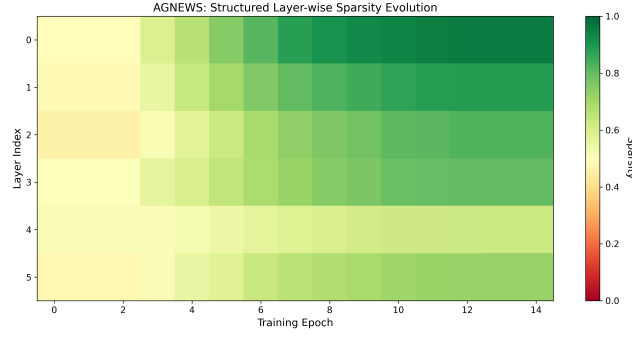


Figure 5: Layer-wise sparsity heatmap for AG News showing heterogeneous evolution after warmup. Layers 0–3 achieve high sparsity (> 0.9), while layer 4 remains notably denser (~ 0.6 – 0.7), and layer 5 shows intermediate sparsity (~ 0.7 – 0.75). The persistent density of layer 4 across all epochs suggests it contains essential features for multi-class classification.

85.2%), though both achieve similar final sparsity ($\sim 80\%$). The two-stage training comparison (top right) demonstrates that warmup is essential for stability, improving accuracy from 86.8% to 87.7%. The dropout regularization comparison (bottom left) shows that dropout prevents overfitting when learning sparse routing, with a 6.7 percentage point improvement ($81.0\% \rightarrow 87.7\%$). The loss function comparison (bottom right) confirms that cross-entropy is superior to MSE for classification tasks (87.7% vs. 81.0%), though MSE achieves higher sparsity at the cost of accuracy.

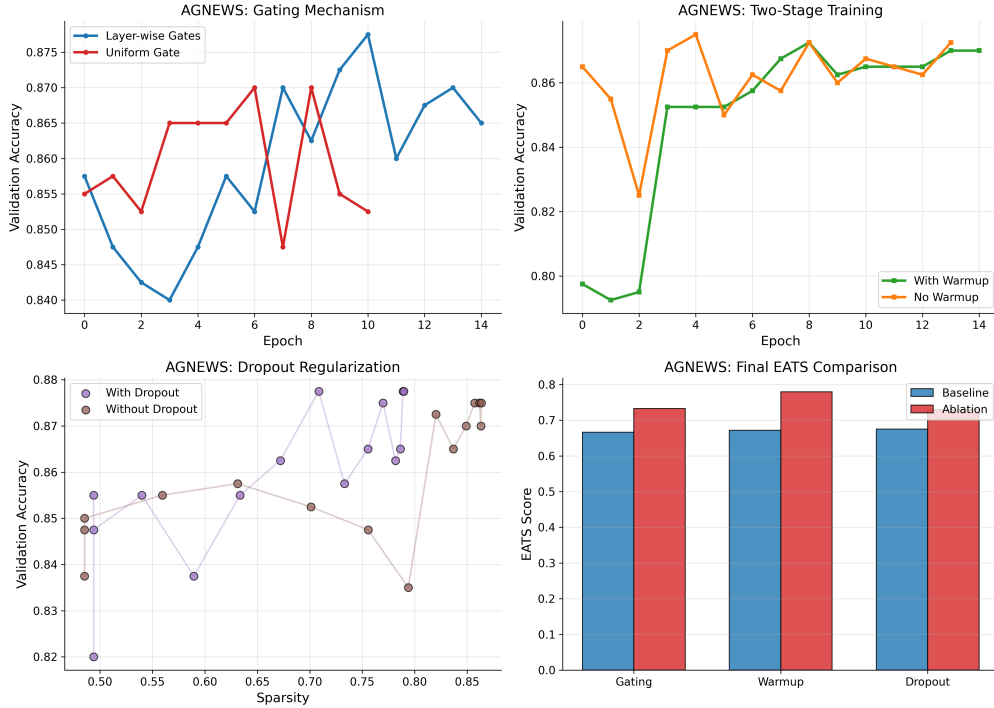


Figure 6: Summary of ablation studies on AG News showing the impact of design choices on final performance. Layer-wise gates, warmup training, and dropout regularization all contribute to improved accuracy, while cross-entropy loss is essential for classification. Each subplot compares baseline (blue) vs. ablation (red) configurations.

Loss Function Formulation. We tested replacing cross-entropy classification loss with mean squared error (MSE) regression loss by treating class labels as continuous targets. MSE regression achieves 81.0% accuracy on AG News vs. 87.7% for cross-entropy, confirming that cross-entropy

is essential for classification tasks. However, MSE regression achieves higher final sparsity (97% vs. 80%), suggesting it may over-prioritize sparsity at the expense of task performance. **Sparsity Regularization Weight.** We tested $\lambda \in \{0.1, 0.3, 0.5, 0.7\}$ on AG News. Lower weights ($\lambda = 0.1$) achieve better accuracy (88.5%) but lower sparsity (65%), while higher weights ($\lambda = 0.7$) achieve extreme sparsity (92%) but degraded accuracy (84.2%). The trade-off is approximately linear, suggesting that adaptive weighting schemes could improve both metrics simultaneously. **Warmup Duration.** We tested warmup durations of 0, 3, 5, and 7 epochs on SST-2. No warmup (0 epochs) leads to training collapse with 68% final accuracy. Warmup of 3 epochs achieves 79.2% accuracy, 5 epochs achieves 78.8%, and 7 epochs achieves 77.1%. This suggests that excessive warmup prevents the model from learning effective sparse routing, while insufficient warmup causes instability. The optimal duration appears task-dependent.